HARSHUL YADAV

AI/ML ENGINEERING | DATA SCIENCE | DEEP LEARNING

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PROFESSIONAL SUMMARY

Third-year B.Tech student in Computer Science and Engineering (Data Science) at USICT, GGSIPU, with AI/ML internship experience at CRIS and a 7.5 CGPA. Builds machine learning and deep learning projects spanning predictive maintenance, audio classification, computer vision, and web-based AI applications. Interested in applied AI, AI-agent security, and data science.

EDUCATION

University School of Information, Communication and Technology (USICT), GGSIPU — New Delhi
B.Tech, Computer Science and Engineering (Data Science) | Expected 2028 | CGPA: 7.5/10

EXPERIENCE

Centre for Railway Information Systems (CRIS) — AI/ML Intern | June–July 2026 (1.5 months)

- Completed an AI/ML internship focused on a railway predictive maintenance project.

SELECTED PROJECTS

Railway Predictive Maintenance System | CRIS internship project

- Combined sensor-based failure prediction, anomaly detection, maintenance prioritization, and railway fastener defect classification.
- Used XGBoost for failure prediction/classification, Isolation Forest for anomaly detection, and EfficientNetB0 transfer learning for fastener images.
- Integrated two FastAPI services with a browser dashboard; reported 85% overall accuracy for binary failure prediction and ~91.4% validation accuracy for fastener classification.

Illegal Logging Detection System with Live Monitoring | 2026

- Built an acoustic AI classifier using ResNet18 transfer learning across seven audio categories; reported 81.2% test accuracy, 83.3% precision, and 0.937 AUC.
- Implemented live microphone monitoring with 5-second windows and 3-sample smoothing, reducing false positives by 60%; built a Gradio interface with waveform, Mel spectrogram, and detection-history views.
- Generated 128-band Mel spectrograms and used four augmentation techniques to expand 147 samples to 735.

Deep Learning-Based Deepfake Detection System | 2025

- Built a ResNeXt-50 detector trained on 60,000+ images, reporting ~95% validation accuracy and inference under two seconds.
- Created a Flask web app with HTML/CSS/JavaScript and OpenCV; implemented Grad-CAM heatmaps to visualize facial regions influencing predictions.

Image-Based Crop Disease Detection Using CNNs

- Studied plant disease classification across crop images using CNNs and transfer learning; the paper describes 54,303 images, 38 disease classes, and 14 crop species.
- Explored ResNet/EfficientNet, image normalization, augmentation, and learning-rate scheduling; project paper reports 99.23% validation accuracy.

AgentGuard — AI Agent Security & Reliability Research | Research in progress

- Reviewing task-aware runtime monitoring for unsafe sequences of individually permitted AI-agent actions; planning a simulated banking environment for evaluation.

AI-Enabled Competency Gap Analysis & Personalized Training Platform | In development

- Developing a platform for competency-gap identification and personalized training for India's Official Statistical System, with AI-assisted quiz generation and planned iGOT Karmayogi integration.

TECHNICAL SKILLS

Programming: Python, JavaScript | **ML:** scikit-learn, XGBoost, Isolation Forest
Deep Learning: PyTorch, TensorFlow/Keras, CNNs, ResNet18, ResNeXt-50, EfficientNetB0, transfer learning
Audio & Vision: torchaudio, librosa, sounddevice, OpenCV, Grad-CAM, Mel spectrograms
Data & Visualization: NumPy, pandas, Matplotlib | **Backend/Web:** FastAPI, Flask, Gradio, HTML, CSS, REST APIs
Tools: Git, Kaggle Notebooks, Visual Studio Code

ACHIEVEMENTS & CERTIFICATION

First Runner-Up — Oblivion Hackathon, NSUT
Machine Learning A-Z: ML, DL, AI with AWS, Python & R — Completed